

Hrithik Raj

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Education

Indraprastha Institute of Information Technology, B.Tech Computer Science and Design 2023 – Present | New Delhi
VV DAV Public School, PCM+CS 2016 – 2023 | New Delhi

Skills

Expertise Area: Data Structures & Algorithms, UI/UX Design, SQL, Frontend Development, Network Analysis, OS, Game Development, Graph Theory, Link Prediction, DBMS, Machine Learning, Routing, Switching.

Programming Language: Python, Java, C++, C, Numpy, HTML/CSS.

Tools and Technologies: VS Code, Figma, Adobe Illustrator, PostgreSQL, Leetcode, Codeforces, Github, IntelliJ, RestAPI, FastAPI.

Professional Experience

Research Intern, Project, intelligent scraping tool, Midas Lab, IIIT Delhi. 08/2025 – 11/2025
• Constructed an automated data pipeline using **Python** and **yt-dlp** to extract audio, subtitles, and **metadata** from 500+ YouTube videos across 12 languages.
• Reduced manual data collection effort by ~85% by automating subtitle parsing (VTT/SRT/JSON), deduplication, and structured **JSON** export.

Projects

Network Analysis link Prediction 🔗, *Machine Learning Developer, Network Analytics Engineer.* 01/2026 – 04/2026
• Benchmarked 5 link prediction algorithms across 3 network topologies, **achieving 0.79 ROC-AUC and 56% Precision@K on real world data.**
• Analyzed 2,900+ edges in Wikipedia voting network using Adamic Adar & Resource Allocation, **outperforming baseline methods by 35%.**

Natural Disaster Management System 🔗, *Database Developer.* 01/2025 – 04/2025
• Engineered a full-stack Disaster Management System with 10+ normalized **PostgreSQL** tables and 7 RESTful API endpoints, reducing data redundancy by ~40%.
• Implemented 3 database triggers enforcing realtime business constraints, improving data integrity and cutting invalid assignments by 100%.

Angry Bird Game Implementation 🔗, *Java Developer, Gameplay Programmer.* 09/2024 – 11/2024
• Developed a **2D Angry Birds-inspired game** using **Java and LibGDX**, implementing OOP-based physics, collision detection, level management, and responsive gameplay mechanics.
• Created and integrated **100% custom UI and game assets** in Adobe Illustrator, optimizing rendering performance for smooth gameplay and scalable graphics across 2D resolutions.

Real time sign language recognition model 🔗, *Machine Learning Developer, Python Developer.* 01/2026 – 04/2026
• Architected real-time sign language recognition system with three modes, achieving accurate detection and low-latency inference.
• Operationalized 40+ gesture recognition models using **MediaPipe** and **scikit-learn** enabling voice output synthesis.

DermaAI 🔗, *Smart Skincare Analysis, Machine Learning Developer, Python Developer.* 02/2026 – 04/2026
• Created a web application using FastAPI, connecting a user-friendly frontend to a Python backend to perform **real time skin analysis.**
• Integrated 3 machine learning models (ResNet50 and EfficientNet-B3) and created custom logic to combine their outputs, **eliminating conflicting predictions for high accuracy.**
• Optimized a personalized product recommendation engine that filters products based on analysis results to match users with the right skincare routine, **reducing bad product matches by 95% .**

Position of Responsibility

E-Cell IIIT Delhi, Coverage Member.
Documented entrepreneurship events through photography, producing engaging media for promotional and social platforms.

Roamsrover IIIT Delhi, Photography Lead.
Led photography team, managed visual content, and enhanced organization branding across digital platforms.

Geek Room Delta, GeeksforGeeks (GFG), Core Community Member.
Fostered community engagement, collaborated with peers, and supported learning initiatives across campus.

Certificates

Edge AI for Microcontrollers 🔗 (University of California, Berkeley.)

Interests and Hobbies

Data Structures, Game Development, UI/UX, Photography, Skating, Film enthusiast.